Dear editors,

Re: "Unsupervised Functional Stage Detection From EEG Using Topological Data Analysis"

Manuscript reference: MBEC-D-25-03498R1

We appreciate the insightful comments and suggestions provided by you and the reviewers, as well as the time and effort devoted to evaluating our manuscript. We have carefully considered the reviewers' comments and addressed them in the revised version of the paper. Below, we provide a point-by-point response to the comments and outline the corresponding revisions made to the manuscript. Where we did not adopt a suggested change, we explain our reasoning below.

[Reviewer #4] The Discussion section could be strengthened by addressing potential applications of shorter-epoch analysis. Standard polysomnography scoring uses 30-second epochs, yet the authors employed 15-second epochs. Shorter epochs have been shown to improve performance in insomnia research and micro-arousal detection. This could be briefly discussed as a potential advantage of the proposed framework for future high-resolution sleep staging or insomnia studies. While standard polysomnographic scoring typically employs 30-second epochs, the use of 15-second epochs in this study may offer advantages for detecting finer-grained transitions. Prior work has demonstrated that shorter epoch lengths can enhance sensitivity in insomnia and sleep fragmentation studies (https://pubmed.ncbi.nlm.nih.gov/20703545/ ; https://pubmed.ncbi.nlm.nih.gov/20703539/). Future investigations could systematically compare epoch durations within the TDA-SDA framework."

[Reviewer #4] Meditation data used primarily 1-second epochs (with limited overlap for one subject). Sleep data used 15-second epochs. The latter deviates from the conventional 30-second AASM/R&K standard for sleep staging. Justification for 15 seconds is not clearly stated in the main text.

[Reviewer #4] Provide explicit justification for the 15-second epoch length in sleep data and compare it to the 30-second standard.

The use of shorter epochs lets us better preserve the temporal precision and granularity of the detected stages, which is indeed a potential additional advantage of the proposed framework that might enable high-resolution EEG studies. We have added a note explaining the rationale in Section 3.2. Additionally, we have now repeated the experiments and report the results with both 15-second and conventional 30-second epoch lengths to explicitly highlight these advantages. The revisions have been incorporated in Section 3.2 and Table 2, and potential applications of these results have been addressed in the Discussion.

[Reviewer #4] Bandpass filtering, ICA artifact removal, and PSD-based epoch rejection are described, but details (e.g., ICA component selection criteria, referencing scheme, exact PSD computation method for rejection) remain limited.

[Reviewer #4] The method used for power spectral density calculation in preprocessing (e.g., Welch's method, multitaper, periodogram) is not specified. Different PSD estimation approaches can significantly affect sensitivity to artifacts and signal quality. Clarify the PSD computation method used for epoch rejection and refer this study on how PSD approaches influence EEG analysis (https://pubmed.ncbi.nlm.nih.gov/18941895/).

[Reviewer #8] Sleep-EDF: The authors mention using the Sleep Cassette Study (SC) subset and discarding the Sleep Telemetry Study (ST). It would be helpful to explicitly state in one sentence why ST was excluded (e.g., "The ST subset was excluded as it contains recordings from patients with known sleep disorders or under medication, whereas we focused on healthy aging").

[Reviewer #8] Meditation Data: In Section 2.1.1, the authors mention that 10-15% of epochs were removed based on PSD deviation. Briefly clarify if this artifact rejection was done before or after the TDA feature extraction, as TDA is highly sensitive to outliers and noise.

We have clarified these details in Section 2.1. The revised manuscript includes the PSD calculation method (multitaper), the justification for excluding the Sleep Telemetry dataset, and an explicit reference to the original study for the specific details of the preprocessing pipeline, including ICA and PSD-based epoch filtering, which was performed before the TDA feature extraction.

[Reviewer #4] Abstract: "supercedes" should be "supersedes".

[Reviewer #5] "supercedes" should be corrected to "supersedes" in the Abstract.

[Reviewer #4] Some sentences in Methods (especially 2.3-2.4) are long and could be simplified.

[Reviewer #5] "mediation" appears to be a typo for "meditation" in the discussion of dimension-5 simplices.

[Reviewer #5] "accross" should be corrected to "across".

[Reviewer #5] The terminology could also be standardized in several places (e.g., using simplices rather than simplexes, where appropriate).

[Reviewer #5] Some expressions would benefit from language polishing for clarity and style, for example phrases such as "via the research of …" or repeated use of "final answer" in methodological descriptions.

[Reviewer #8] Abstract: The phrase "several thousand features" is used. Given that Section 3.1 mentions "almost 20000 topological features", it might be more accurate to say "tens of thousands of features" in the abstract to emphasize the need for the QSDA selection framework.

[Reviewer #8] Figure 4 Caption: "Number of occurrences of each brain region in the dimension-5 representative cocycles..." Ensure that the abbreviations (FP, LF, RF, LT) are defined either in the caption or clearly in the main text where they first appear.

We have corrected the typos and incorporated the valuable revisions as requested. We have also proofread the entire paper again and implemented minor structural and stylistic improvements throughout the manuscript.

[Reviewer #5] The paper does not clearly state whether the observed performance improvement results from the proposed topological representation itself or from the substantially expanded feature space followed by feature selection.

[Reviewer #6] The feature engineering pipeline is extremely complex, generating tens of thousands of features through multiple TDA representations, filtering procedures, dimensions, amplitudes, landscapes, silhouettes, and statistical summaries. Although QSDA is proposed to reduce

dimensionality, the risk of overfitting and selection bias remains substantial. Since feature selection is optimized using the same SDA framework that is later evaluated, there is a possibility that the procedure is favoring features specifically aligned with the clustering objective rather than identifying genuinely informative physiological characteristics. Additional validation on completely independent datasets or through nested evaluation procedures would strengthen confidence in the robustness of the proposed framework.

QSDA has been introduced specifically to address this issue by reducing the topological feature set to a scale comparable to spectral features. Further, both pipelines have employed an identical PCA reduction to 15 components before SDA, which is an additional step to control dimensionality. Nonetheless, we agree that potential bias resulting from the expanded feature space indeed remains a reasonable concern that requires further research.

The circularity of QSDA also introduces the risk of selection bias. However, we believe the impact to be largely limited, since the algorithm relies on individual feature scoring rather than joint optimization. The consistent agreement with information value, an independent criterion, and extensive validation techniques (noisy and shuffled recordings, Mann–Whitney statistical tests, Kruskal–Wallis spectral validation) also support the robustness of the framework.

Regardless, we acknowledge that a nested evaluation would provide stronger formal guarantees and have explicitly noted both remarks in the Discussion as limitations and directions for future work.

[Reviewer #5] The manuscript does not provide sufficient analysis regarding the sensitivity of the proposed method to key TDA hyperparameters. For improving the reproducibility and methodological clarity, the authors are encouraged to explicitly report and discuss the impact of varying key parameters such as filtration settings, embedding choices and persistence thresholds on the final classification performance within the manuscript.

[Reviewer #6] Several methodological choices require stronger justification. The selection of 15 PCA components is only partially motivated by consistency with previous work, and a more principled criterion would be preferable. Similarly, the thresholds used in QSDA, the exclusion of features with fewer than 200 unique values, and the filtering strategies applied to persistence diagrams appear somewhat heuristic. Sensitivity analyses demonstrating that the main conclusions remain stable under alternative parameter choices would improve methodological rigor.

We have explored the sensitivity of the framework to four primary hyperparameters noted by the reviewers (the number of PCA components, the minimum number of unique values required for the feature to be considered in QSDA, the number of features selected in QSDA, the fraction of points removed from persistence diagrams) and report the results in Supplementary Materials B, which is referenced in Section 3.1. We have also clarified the rationale for choices of QSDA thresholds and the number of PCA components in the relevant sections of the revised manuscript.

[Reviewer #8] Takens Embedding: In Section 2.3, the authors mention using general hyperparameter selection methods for Takens embedding (dimensionality and time delays). Please cite the specific methods or software packages used to calculate these (e.g., False Nearest Neighbors, Mutual Information) if not already cited.

We now explicitly state in Section 2.3 that the time delay was selected using mutual information [1], while the embedding dimension was determined using the false-nearest-neighbors method [2]. We also added the name of the software implementation used throughout the study (giotto-tda [3]), including the hyperparameter selection methods for Takens embedding, at the end of Section 2.4.

[1] Fraser AM, Swinney HL (1986) Independent coordinates for strange attractors from mutual information. Phys Rev A 33:1134–1140. https://doi.org/10.1103/PhysRevA.33.1134

[2] Kennel MB, Brown R, Abarbanel HDI (1992) Determining embedding dimension for phase-space reconstruction using a geometrical construction. Phys Rev A 45:3403–3411. https://doi.org/10.1103/PhysRevA.45.3403

[3] Tauzin G, Lupo U, Tunstall L, et al (2021) giotto-tda: A topological data analysis toolkit for machine learning and data exploration. URL https://arxiv.org/abs/2004.02551, arXiv:2004.02551

[Reviewer #8] Filtration Parameter: For the Vietoris-Rips complexes, what was the maximum filtration parameter or the maximum simplex dimension used? This is a crucial hyperparameter in TDA that heavily influences the resulting persistence diagrams.

Although this was previously available in Supplementary Materials B, we agree that these are crucial hyperparameters that might significantly impact reproducibility. We have therefore explicitly specified the maximum simplex dimension used for all three branches of the algorithm in Section 2.4 of the main text, and noted that no fixed maximum filtration value was imposed, with the filtration range determined automatically from the distance matrix by the persistence implementation.

[Reviewer #5] Was PCA fitted independently to each training layer, or was it computed on the entire dataset before evaluation? It is still unclear in the pipeline that should be directly clear in the manuscript.

PCA was fitted independently to each subject's feature space, which is an intentional design choice to preserve the unsupervised nature of EEG segmentation. We have clarified this detail in Section 2.3.

[Reviewer #5] The link between the identified dimension-5 simplices and limbic system activity appears speculative. In this part, for supporting this claim, additional evidence or can be presented.

[Reviewer #6] The physiological interpretation of the findings remains speculative. For example, the discussion linking dimension-5 simplices to limbic system activity and meditation transitions is intriguing but currently lacks sufficient neuroscientific evidence. The presented cocycle analysis demonstrates spatial associations but does not establish a causal or functional relationship. These interpretations should be considerably softened or supported by additional analyses and references from neurophysiological literature.

As noted by the reviewer, the cocycle analysis establishes only a spatial association and does not imply a direct functional relationship. We have softened the claims regarding this observation and explicitly acknowledged the physiological validation of this result as a promising direction for future work.

[Reviewer #6] Another concern relates to the ground truth problem in the meditation dataset. Since no objective annotations are available, the manuscript relies heavily on internal clustering metrics and comparisons with previous SDA outputs. High silhouette scores do not necessarily imply physiologically meaningful states. Consequently, conclusions regarding the existence of previously unknown functional stages should be presented more cautiously. The sleep dataset provides stronger validation because expert labels exist, but even there the matching procedure between predicted and expert-defined stages introduces flexibility that may inflate performance estimates.

[Reviewer #7] The discussion section does not adequately interpret the observed improvements or explain the underlying reasons for the performance differences. As a result, the claimed superiority of the proposed method is not sufficiently substantiated.

To support the physiological relevance of the findings and address the absent ground truth in the meditation dataset, we have introduced additional techniques to assess whether detected boundaries reflect the spectral characteristics of the EEG: the Kruskal–Wallis and Mann–Whitney tests on the band spectral power. As suggested by Table 1 and Fig. 7 of the revised manuscript, the boundaries in fact meaningfully relate to the spectral density for both meditation and sleep datasets. However, a physiological validation of the findings remains a crucial direction for future work.

Concerning the sleep dataset, we agree that selecting the best-fitting subset of expert boundaries may lead to optimistic FMI estimates. However, the comparison is intended to assess structural correspondence rather than provide an unbiased estimate of conventional sleep-staging accuracy. We have acknowledged this issue as a limitation in Section 3.2.

[Reviewer #6] A major concern is the repeated claim that topological features "outperform," "supersede," or are "superior" to traditional spectral features. While the authors report higher silhouette scores and FMI values, the statistical evidence remains incomplete. The argument based on confidence interval separation and sigma differences is not equivalent to a formal statistical comparison between competing methods. Direct hypothesis testing across subjects should be performed, including appropriate paired statistical tests and effect size estimates, to support claims of superiority. Without such analyses, the conclusions appear overstated.

To provide a more formal statistical analysis of the results, we now report direct hypothesis testing with paired t p-values and effect sizes in Table 2. We have also softened the superiority claims throughout the manuscript to acknowledge the limited physiological evidence available.

[Reviewer #8] Introduction: The authors mention "somnology and epilepsy detection" as valuable continuous processes. While the paper covers somnology (sleep), epilepsy is not tested. Consider slightly rephrasing to ensure the scope of the paper isn't overstated, or frame it as "tasks such as somnology and, potentially, epilepsy detection."

[Reviewer #6] Finally, the manuscript occasionally overstates its conclusions relative to the presented evidence. Statements suggesting that TDA reveals new physiological patterns, supersedes spectral features, or provides a trustworthy alternative in general EEG analysis are stronger than what the current experiments support. The results demonstrate promising performance on two datasets and justify further investigation, but broader claims should be moderated until validated across additional EEG paradigms and independent cohorts.

[Reviewer #8] In the Discussion section, the authors state: "In practice, one should combine the techniques to capture all aspects of the data and obtain a robust outcome." However, in the Sleep-EDF results (Table 1), the "Combined" feature set actually performs slightly worse than the "Topological" feature set alone in terms of both Adapted Silhouette (0.227 vs 0.228) and FMI (0.907 vs 0.911). Please reconcile this statement. It appears that for the sleep dataset, the topological features alone are sufficient and adding spectral features introduces slight noise or redundancy (curse of dimensionality). The authors should nuance their conclusion to reflect that while combining features can be beneficial (as seen in some meditation subjects), topological features alone can supersede traditional ones without needing combination.

We have considerably softened and nuanced the conclusions throughout the manuscript, including the Abstract and the Discussion section. The superiority claims indeed require further physiological validation, which remains a direction for future studies. The imprecise statement about the combined feature space in the Discussion has been reconsidered to better recognize the findings from the Sleep-EDF dataset.

[Reviewer #6] The manuscript would also benefit from a more rigorous discussion of computational scalability. Although runtime comparisons are presented, the practical feasibility of generating approximately 20,000 topological features for larger datasets or real-time applications remains unclear. Memory requirements, computational complexity, and reproducibility considerations should be discussed in greater detail, particularly given that TDA methods are often computationally demanding.

[Reviewer #8] While the authors provide a relative comparison of running times in Figure 7, it is difficult to gauge the absolute computational burden. TDA (specifically calculating Vietoris-Rips complexes and persistent homology) is notoriously computationally expensive. Action required: Please provide the absolute processing time (e.g., in minutes or hours) required to process a single subject's recording (e.g., a full night of sleep or a full meditation session) using the proposed TDA pipeline on standard hardware. Discuss briefly whether this computational cost is a barrier for real-time or large-scale clinical applications, and mention potential avenues for accelerating the TDA calculations (e.g., approximate algorithms, parallelization).

[Reviewer #10] The computational cost of each stage is not analyzed. Since persistent homology is computationally demanding, scalability may become an important concern for longer EEG recordings or high-density EEG systems. Include a computational complexity analysis (Big-O notation if possible), together with average runtime and memory consumption for the principal stages of the algorithm.

We have expanded the discussion of computational scalability and now report the absolute running times for full meditation and sleep EEG. Additionally, we explicitly acknowledge persistent homology as the primary bottleneck and outline the optimizations that can be introduced to reduce the computational burden.

Unfortunately, because the end-to-end runtime depends on several branches and implementation-specific persistence algorithms, a single informative closed-form Big-O expression is difficult to provide. We therefore report empirical runtimes and memory use and discuss the combinatorial worst-case growth of Vietoris–Rips complex construction.

[Reviewer #7] The manuscript lacks a comprehensive comparison with relevant state-of-the-art EEG feature extraction methods, particularly existing TDA-based and spectral-based approaches.

[Reviewer #7] Positioning of the contribution within the existing literature is weak, making it difficult to assess true novelty.

A new paragraph was added to the Introduction after related work that explicitly positions the contribution within the existing literature: current TDA work is predominantly supervised, whereas spectral methods dominate the unsupervised domain. To the best of our knowledge, among the studies reviewed, no prior work has evaluated TDA within an unsupervised, annotation-free framework or compared it directly to spectral methods in these settings. Additionally, the Discussion and Conclusion sections were adjusted to highlight the contribution and the summary comparison with the state-of-the-art spectral method explicitly.

[Reviewer #10] It is recommended that the authors strengthen the paper by incorporating several recent advanced studies (DOIs: https://doi.org/10.1016/j.cosrev.2025.100792, https://doi.org/10.1016/j.knosys.2025.114975, https://doi.org/10.1016/j.cosrev.2026.100974, https://doi.org/10.1109/TKDE.2026.3680286)

After careful consideration, we decided to add the two manuscripts describing the applications of transformers and diffusion learning to several clustering problems. We have incorporated the corresponding references in the second paragraph of the Introduction.

The methodological focus of the remaining two papers is not directly connected to EEG segmentation, neurophysiological TDA, or biomedical spectral feature extraction. We have therefore not included them in the revised paper, as doing so would not strengthen the specific literature context of the present study.

[Reviewer #8] The authors state that raw data will be made available upon request. However, the core contribution of this paper is the TDA feature extraction pipeline and the QSDA feature selection algorithm. To maximize the impact and reproducibility of this work, I strongly encourage the authors to make the source code for the TDA pipeline, QSDA, and the SDA implementation publicly available (e.g., via a GitHub repository) and include the link in the Data Availability statement.

As requested by the reviewer, we made the source code publicly available on GitHub and included a link in the Data Availability statement.